Mock Interview: Cassandra

Test your knowledge of Cassandra's architecture by answering the following questions.

Quiz ?

Statement

How does Cassandra scale?

Answer

Cassandra uses **consistent hashing**, and with the help of **Vnodes**, adding nodes to the cluster is quite easy. When a new node is added, it receives many Vnodes from the existing nodes to maintain a balanced cluster.

Hide Answer

Quiz ?

Statement

How does Cassandra ensure fault-tolerance and reliability of data?

Answer

Cassandra replicates data to multiple nodes for reliability and fault-tolerance.

Hide Answer

Quiz ?

Statement

What write consistency levels does Cassandra provide to its clients?

Answer

For write operations, the consistency level specifies how many replica nodes must respond for the write to be reported as successful to the client. Cassandra offers different write consistency levels:**One**, **Two**, **Three**, **Quorum**, **ALL**, **Local_quoram**, **Each_quorum**, and **Any**.

Hide Answer

Quiz ?

Statement

What read consistency levels does Cassandra provide to its clients?

Answer

The consistency level for read queries specifies how many replica nodes must respond to a read request before returning the data. For example, for a read request with a consistency level of quorum and replication factor of three, the coordinator waits for successful replies from at least two nodes. For read requests, Cassandra has the same consistency levels as that of write operations except **Each_quorum**.

Hide Answer

Quiz ?

Statement

How does Cassandra ensure strong data consistency?

Answer

To achieve strong consistency in Cassandra:$R + W > RF$ => strong consistency. In this equation, R, W, and RF are the read replica count, the write replica count, and the replication factor respectively. All client 'reads' will see the most recent 'write' in this scenario, and we will have strong consistency.

Hide Answer

Quiz ?

Statement

How does Cassandra ensure durability?

Answer

Cassandra ensures durability by replicating data across multiple replicas, as well as storing them on the disk.

Hide Answer

Quiz ?

Statement

How does Cassandra manage high availability?

Answer

Each piece of data is replicated to multiple nodes to ensure high availability. The replication factor is the number of nodes that will receive the copy of the same data. This means, if a cluster has a replication factor of 3, each row will be stored on three different nodes. In this case, the data will still be available even if two nodes fail.

Hide Answer

Quiz ?

Statement

What is the use of Bloom Filter in Cassandra?

Answer

On a read request, before doing any disk I/O, Bloom filter is used to check whether the requested data exist in an SSTable.

Hide Answer

Quiz ?

Statement

How is an SStable stored on the disk?

Answer

SStable consist of mainly two files. Index file (Bloom filter & Key offset pairs) & Data file (actual columns data).

Hide Answer

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